

A Game Theory Based Collusion Aware Auction Framework for Agentic Compute Markets*

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Abstract—This paper studies agentic compute markets where autonomous bidders may collude. We compare a baseline second price auction to an improved design with collusion aware bid enforcement and machine learning based round level monitoring. Simulations use many agents and repeated rounds with multi trial aggregation. We report average payment and welfare, detector performance, and include comparison plots for payment and welfare versus round.

Index Terms—game theory, vickrey auction, collusion detection, agentic markets, compute marketplace, adaptive enforcement

I. INTRODUCTION

Market style allocation of compute for AI workloads increasingly relies on software agents that bid on behalf of users. Coordinated bidding can depress prices and harm efficiency. This work presents a reproducible simulation that compares baseline clearing to an improved pipeline with adaptive enforcement while preserving a second price payment rule.

II. METHODOLOGY (SUMMARY)

Agents draw valuations from a normalized hardware table. Each round, honest and collusive bidding modes produce a bid vector. Round level features summarize bid dispersion and temporal structure. The improved system adjusts bids before clearing using an online risk score. Outcomes record second price payment and a welfare proxy tied to the winning bid. The implementation exports figures under a graphs folder.

III. EXPERIMENTAL SETUP

Default protocol uses one hundred fifty agents, two hundred rounds per trial, and twenty seeded trials. The dataset path is set in the project configuration file. Compile this paper from the project root so relative paths to PNG files resolve.

Identify applicable funding agency here. If none, delete this.

IV. RESULTS

A. Aggregate metrics

Table I summarizes trial level means and standard deviations from the same pipeline that generates Fig. 1 and Fig. 2. Update the numeric cells after each full run of `python main.py` if you change parameters.

TABLE I
MULTI TRIAL SIMULATION SUMMARY (TWENTY TRIALS, STRATIFIED
DETECTOR EVALUATION PER TRIAL).

Metric	Value
Basic mean payment	3.8573 $\pm$ 0.0009
Improved mean payment	4.6247 $\pm$ 0.0190
Basic mean welfare	3.8711 $\pm$ 0.0013
Improved mean welfare	4.6399 $\pm$ 0.0190
Payment gain (Improved vs Basic)	19.90%
Welfare gain (Improved vs Basic)	19.86%
Accuracy	0.9200 $\pm$ 0.0267
ROC-AUC	0.9284 $\pm$ 0.0291
Precision	0.9224 $\pm$ 0.0364
Recall	0.9146 $\pm$ 0.0465
FPR	0.0773 $\pm$ 0.0382
FNR	0.0854 $\pm$ 0.0465

B. Payment comparison

Fig. 1 plots second price payment versus round for the Basic and Improved systems. The horizontal axis is Round; the vertical axis is Payment. The Improved series lies above Basic on average, consistent with Table I.

C. Welfare comparison

Fig. 2 plots the welfare proxy versus round for both systems. Axes are Round and Welfare. Patterns mirror the payment

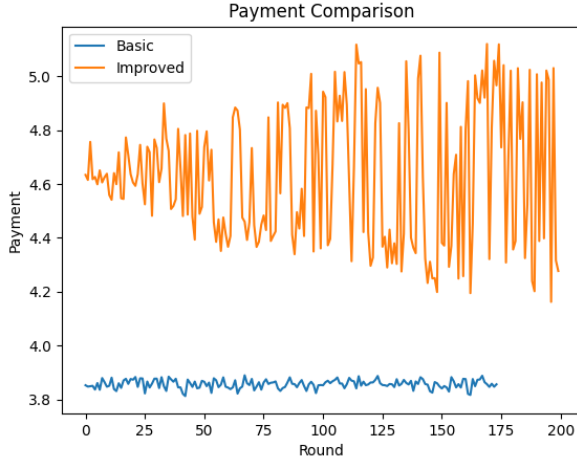


Fig. 1. Payment comparison: Basic and Improved second price payments over rounds (axes: Round, Payment).

figure because both quantities come from the same auction clearing each round.

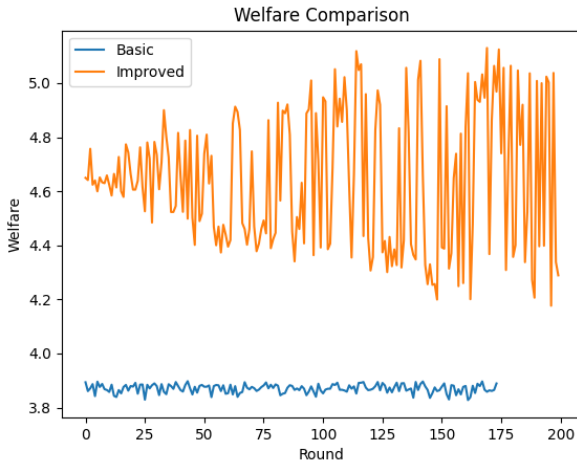


Fig. 2. Welfare comparison: Basic and Improved welfare proxy over rounds (axes: Round, Welfare).

V. DISCUSSION

Detector metrics in Table I reflect a synthetic task and should not be read as guaranteed real world performance. Cross dataset tests and noisier labels are recommended before deployment claims.

VI. CONCLUSION

We presented a game theory motivated collusion aware auction simulation with clear baseline versus improved comparison, quantitative aggregation in Table I, and visual evidence in Fig. 1 and Fig. 2.

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